

GENERAL SPECIFICATION G2.28

Spare Parts, Tools and Test Equipment

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G2.28 SPARE PARTS, TOOLS AND TEST EQUIPMENT

G2.28.1 Overview

- (a) The Contractor shall provide for Board's use the following as defined in relevant Bills of Quantities:
 - (i) Critical spare parts; and
 - (ii) Additional spare parts
- (b) The Contractor shall provide for the Board's use, any special tools and test equipment needed for routine maintenance or overhaul of Plant and equipment for a period of 2 years operation after the expiry of the DLP.
- (c) Spare parts, tools, test equipment and consumable items needed for the equipment provided under the Contract shall be new and shall be procured only from the manufacturers of the Plant and equipment provided.
- (d) Spare parts, tools, and test equipment shall be packed separate from the main Plant and equipment in packages or containers designed to preserve and protect the spares from the effects of long term storage under the ambient conditions prevailing in Singapore and at Site and the TWRP. Any items which cannot be packed in this way must be protected from corrosion by applying temporary protective coatings and shielded from mechanical damage. All items shall be clearly labelled with brief descriptions and part numbers and shall be cross-referenced to the approved spare parts list.
- (e) All spare parts, tools and test equipment shall be delivered to the Site prior to the issue of the Certificate of Substantial Completion. All spare parts, tools and test equipment shall be stored strictly in compliance with manufacturers' recommendations.
- (f) All rotating equipment, when in storage, shall be manually rotated from time to time to prevent flattening of bearings.
- (g) All electric motors shall be stored with anti-condensation heaters energised at all times.
- (h) All crates, containers or other packages are liable to be opened for such examination as the S.O. may require and the packing shall be designed to facilitate the opening and subsequent re-packing.
- (i) The Contractor shall furnish the names and addresses of firms within Singapore that maintain technical service representatives for each foreign-made item or component incorporated into the Works.

G2.28.2 Spare Parts

- (a) The listing of all spare parts shall contain a description of the spare parts, ordering reference numbers with parts identified on equipment drawings, manufacturer details including contact information, part number, lead time and quantities.

- (b) The Contractor shall not use any of the spare parts for construction, commissioning, for rectification of defects or during the Operations and Maintenance periods without the written permission of the S.O.. All parts so used shall be replaced free of charge by the Contractor prior to hand over of the Works to the Board.

G2.28.3 Critical Spare Parts

- (a) Critical spare parts are those, which are deemed necessary by the manufacturer of any item of Plant/equipment to support continuous operation under all envisaged operating conditions for two (2) years after handover to the Board.
- (b) Critical spare parts shall be as described in writing in the manufacturer's operation and maintenance instructions for the item of Plant/equipment.

G2.28.4 Additional Spare Parts

- (a) Additional spare parts are those listed in relevant Bill of Quantities. These are to be priced at the time of Tender. The Board reserves the right to procure none, some or all of the additional spare parts for the lump sum prices given at the time of Tender.

G2.28.5 Special Tools and Test Equipment

- (a) The Contractor shall provide two sets of all the special tools necessary for the dismantling and repair of the Plant or equipment unless otherwise specified.
- (b) All special tools and test equipment listing shall contain a description of the special tools/ test equipment, ordering reference numbers, information on the use of the tools/ equipment, manufacturer details including contact information, part number, lead time, quantities, unit price and the total price.
- (c) Special tools and test equipment shall not be used for the erection of the equipment being supplied and shall be handed over to the Board in a new and unused condition. If the Contractor requires the special tools, he shall make his own arrangements. At the discretion of the S.O., the Contractor may be called upon to demonstrate the effectiveness of any special tool.
- (d) Special tools and test equipment shall be contained in a suitable hardwood, floor mounted cabinet clearly marked or labelled with its description. Each tool shall be identified, and a list of tools shall be affixed to the inside of the cabinet door in a document holder.
- (e) Each set of tools shall be supplied with the equipment with which it is associated.

END OF SECTION